Amirishetti Harshavardhan

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Work Experience

Software Development Engineer - 1 July 2024 – Current

Kotak Mahindra Bank Ltd. *Bangalore, India*

- Designed and led end-to-end development of **DataContract**, a centralized governance platform managing **100+ data contracts across 15+ stakeholder systems**, featuring version-controlled workflows, audit trails, and a maker-checker approval model that reduced manual coordination by **60%**.
- Enabled real-time schema change notifications and batch enforcement on **ETL data (20M+ records/day)** using AWS EventBridge, SNS, and AWS Batch, scaling adoption to 30+ internal teams.
- Scaled the platform with custom integrations to Kotak internal systems, enabling seamless governance onboarding for internal Kotak teams and boosting schema **compliance rate by 75%** in the first quarter of deployment.
- Designed and developed DataVigil, a scalable data quality platform to **validate 80K+** enterprise tables across Sybase and Redshift using **Apache Spark on AWS EMR.**
- Built end-to-end pipelines for rule-based data validation, with **Spark submit orchestration via AWS MWAA**, DynamoDB-based result storage, and in-memory joins for multi-table checks.
- Independently developed both backend (Java Spring Boot microservice) and processing layer (Python Spark framework), enabling dynamic rule execution and automated governance insights.
- Engineered AWS DynamoDB-based result storage architecture to handle high-throughput DQ results ingestion and querying.
- Led end-to-end implementation of Single Sign-On (SSO) feature for Kotak Dataverse application, integrating with the central IDAM platform. Incorporated the Dataverse application with the IDAM platform and configured the firewall whitelisting to secure communication between Dataverse and IDAM.
- Implemented OIDC (OpenID Connect) authentication protocol in the application middleware (Java Play Framework) to authorize all incoming API requests. Developed and integrated in-app and external login / log-out functionality, managing JWT tokens from IDAM server and exchanging them for user profile data from the Kotak's AD.
- Implemented role-based SSO login with custom mapping for user access control. Managed user session duration using session cookies, handling access, and refresh tokens for secure user credential storage. **This improved token management efficiency by 30%**, reducing the load on the IDAM platform during peak usage times.
- Utilized Ory Hydra as an OIDC provider for provisioning tokens, client ID, and secret generation. Developed complex log-out functionality that includes token introspection, revocation, and invalidation to ensure secure session management.
- Configured Horizontal Pod Autoscaler (HPA) and Vertical Pod Autoscaler (VPA) to dynamically manage resources, ensuring optimal scaling and performance of the application.
- Set up Kong as an API gateway using Kong Ingress Controller to manage and route traffic efficiently in a Kubernetes environment. Integrated Kong Rate Limiter to protect APIs by limiting request rates, preventing DDOS attacks, and overloading of services.

Software Engineer April 2024 – June 2024

PTCG India Pvt. Ltd. *Hyderabad, India*

- Developed and implemented the backend of a Customer Relationship Management (CRM) system from scratch using Java and Spring Boot. Designed database tables, schema, and system design to ensure efficient data management and scalability.
- Implemented key features such as SMS and voice call functionalities within the CRM application to enhance communication capabilities. Developed modules for importing data from Excel and CSV files, improving data integration and usability.
- Designed and implemented role-based access control (RBAC) mechanisms to ensure secure user management and data protection. The application was deployed in Azure and the monitoring was performed using Azure Lens, ensuring high availability and performance.

Software Engineer June 2022 – June 2023

Undostres *Mexico City, Mexico*

- Integrated the PayPal payment feature within the Undostres app, orchestrating tokenization and bill creation processes during checkout.
- Engineered an optimized flow using PayPal REST APIs,resulting in a significant reduction in payment processing time from **1.5 minutes to 10 seconds**.
- Constructed OTP authentication functionality for credit card, login, and registration workflows using various SDKs. Implemented the management of user feedback for credit card requests and introduced a multi-device OTP rejection feature for added security.
- Assumed the role of a Full-stack Developer within the CS team and systematically addressed and mitigated the referral abuse challenge. Instrumental in achieving a **35% reduction in referral abuse** through the implementation of targeted solutions.
- Spearheaded the automation of gender and date of birth (DOB) storage for Google Login, enhancing efficiency in user data management. Designed and executed an optimized database schema tailored for the storage of intricate user registration and login details.

Software Engineer Intern Jan 2022 – April 2022

Undostres *Mexico City, Mexico*

- Developed an automated pipeline for streamlined movie ticket bookings, integrating with diverse ticket booking APIs. Implemented file retrieval from AWS S3 and displayed in the transaction history alongside redeem codes.
- Revamped user feedback procedures for credit card requests through the deployment of an automated mailing system. This optimization facilitated rapid responses to customer queries,resulting in a substantial **30% improvement in customer satisfaction**.
- Engineered a robust OTP authentication feature for seamless credit card, login, and registration processes by integrating Firebase and Twilio SDKs; **resulting in a 45% reduction** in fraudulent activities and enhanced user security.

Achievements

- Recognized as one of the top 4 SDEs among a team of 15 and was awarded for delivering key business-critical features ahead of schedule. Spearheaded the completion of high impact features initially slated for Phase 1, delivering them in Phase 0, securing the AppSec sign-off, and facilitating a smooth production launch ahead of time.
- Attained an All India Rank of 5613 among 2 million applicants in JEE-MAINS 2018, showcasing exceptional skills and standing out among a competitive pool.
- Achieved a LeetCode rating of 1747, resulting in top 1000 globally, demonstrating strong coding and problem-solving skills.
- Secured AIR of 11 and 21 among 10000 participants consecutively for 2 years in SICHO.
- Secured the position of Runner-up in the INTER NIT Volleyball Championship, showcasing skill and teamwork on a national level.

Technical Skills

Languages: Java, Python, C++, PHP, Javascript

Frameworks & Protocols: SpringBoot, Django, Hibernate, Flask, ODCS, Auth2, OIDC.

Databases: MYSQL, PostgreSQL, AWS RDS, DynamoDB, MongoDB, Redis.

Technologies: Airflow, Apache-spark, Kubernetes, AWS (EKS, EMR, S3, ECR, CloudWatch, Batch, MWAA), Docker, Kong API Gateway, Terraform, Helm, Azure CI/CD, Podman, Firebase, REST APIs, Postman, Node.js, GitHub, Azure lens.

Relevant Coursework: Data Structures and Algorithms (C++), Design & Analysis of Object Oriented Programming, Advanced Web Programming, Database & Management Systems, Operating System, Computer Networks.

Education

Visvesvaraya National Institute of Technology, Nagpur GPA: 7.55/10.00

Bachelor of Technology in Computer Science and Engineering *June 2018 - June 2022*